

How do feedback processes from AGNs influence the star formation of host galaxies?

Chemical Evolution of Galaxies - ASP 5556

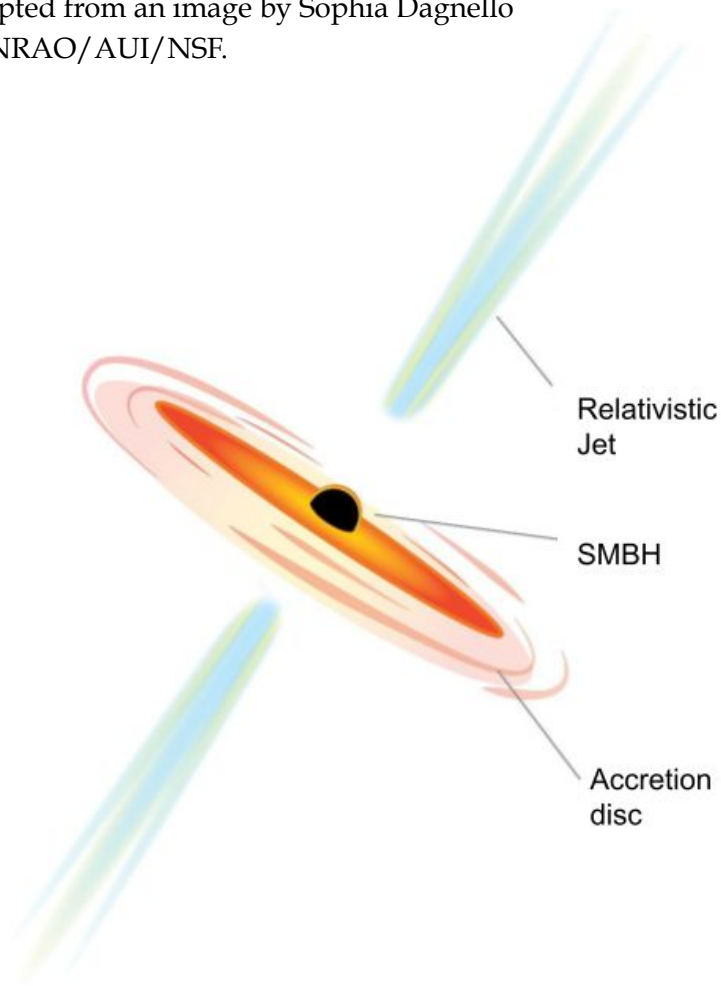
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Adapted from an image by Sophia Dagnello
for NRAO/ AUI/NSF.



An Active galactic nuclei (AGN) is a compact region at the center of a galaxy that emits a significant amount of energy across the electromagnetic spectrum (not produced by stars).

AGNs are one of the key components in understanding galaxy formation and evolution as they release large amount of energy via gas outflows or radio jets, affecting star formation in galaxies as well as the properties of intergalactic medium (Fabian 2012; Harrison 2017).

Forms of Feedback - Begelman (2003)

Positive AGN Feedback

Radiation Pressure: Radiation from the AGN exerts pressure on the surrounding gas.

Jet-Induced SF: Jets interact with the ISM in the galaxy.

Negative AGN Feedback

Radiative Heating: High-energy radiation from the AGN heats the surrounding gas.

Kinetic Energy from Jets and Outflows: Winds and jets inject kinetic energy into the surrounding medium.

Energetic Particles: Particles accelerated by the AGN, such as cosmic rays, interact with the ISM.

Positive feedback mechanisms involve the compression of gas that triggers SF, whereas negative feedback mechanisms involve heating and expelling gas to prevent SF.

Feedback Modes

AGN feedback is theoretically proposed to come in one of two principle forms:

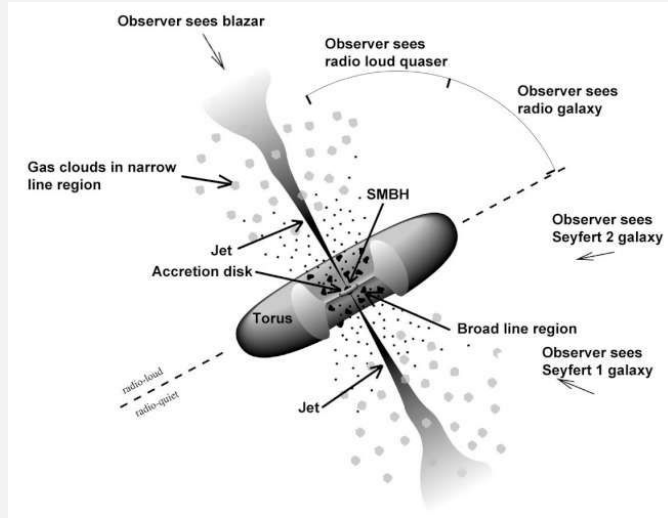
- **Quasar Mode** (e.g., Hopkins et al. 2006; Hopkins et al. 2008; Nesvadba et al. 2008; Feruglio et al. 2010; Maiolino et al. 2012):

Quasar mode feedback describes the processes by which high luminosity AGN trigger galactic winds which can lead to outflows

- **Preventative Mode** (e.g., Bower et al. 2006, 2008; Croton et al. 2006; Fabian 2012):

Preventative mode feedback refers to the heating of the CGM by low luminosity AGN, potentially via relativistic jets or kinetic feedback.

The AGN feedback role on the star formation depends on the AGN type the galaxy hosts.



Radio-Loud AGN

- Radiatively Efficient: Radio-loud QSOs, HERGs.
- Radiatively Inefficient: LERGs.

Radio-Quiet AGN

- Seyfert galaxies and radio-quiet QSOs.

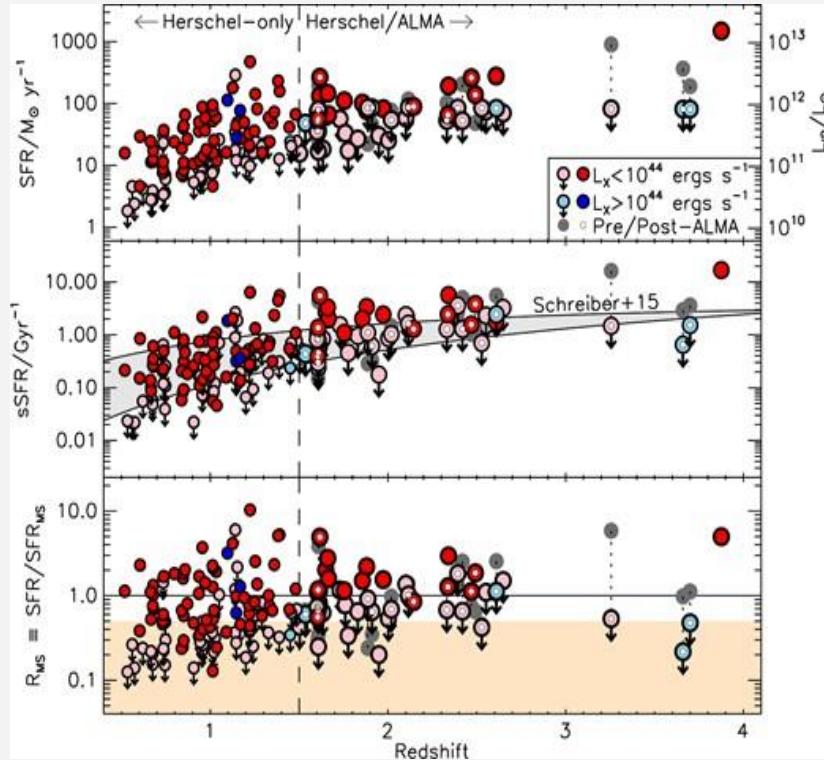
Observational Evidence

- Previous studies have found that AGN add additional complexity to the regulatory processes of SF when compared to non-active galaxies of comparable mass.
- Star-forming galaxies fall on a tight correlation (~ 0.2 intrinsic scatter; Speagle et al. 2014) between SFR and M^* , known as the main-sequence (Brinchmann et al. 2004)

For example, Mullaney et al. (2015) and Shimizu et al. (2015) have found that the SFRs in X-ray-selected AGN host galaxies have more suppressed SFRs than non-active galaxies of similar mass, whereas Shin et al. (2019) found AGN-driven outflows enhancing star formation in a given region of Seyfert 2 galaxy.

Negative Feedback

Observational Evidence - Mullaney et al. (2015)



Using deep *Herschel* and ALMA observations, Mullaney et al. (2015) investigated the SFR distributions of X-ray-selected AGN host galaxies at $0.5 < z < 1.5$ and $1.5 < z < 4$, comparing them to that of normal, star-forming MS galaxies

$$R_{\text{MS}} \equiv \text{SFR}/\text{SFR}_{\text{MS}}$$

Findings: Between 34 and 55 per cent (dependent on upper limits) of AGNs in our combined (i.e. low- z +high- z) sample lie within this shaded region ($R_{\text{MS}} < 0.5$), compared to $\approx 15\%$ of MS galaxies.

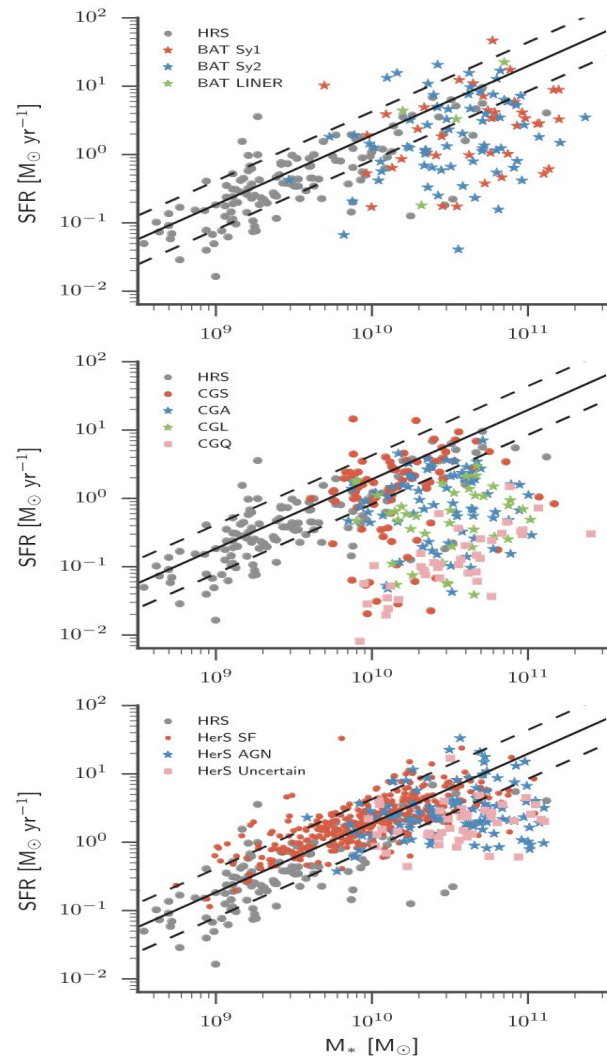
Negative Feedback

Observational Evidence - Shimizu et al. (2015)

Shimizu et al (2015) investigates the position of ultra-hard X-ray AGNs relative to the MS of star-forming galaxies.

Samples: non-AGN SF galaxies (Herschel Reference Survey)/ AGN host galaxies selected from the Swift/BAT (Burst Alert Telescope) catalogue, SF and AGN host galaxies (COLD GASS).

Findings: A significant portion of BAT AGN exhibit lower specific star formation rates (sSFR) compared to non-AGN galaxies. BAT AGNs share similarities with COLD GASS AGNs. Both are positioned between star-forming galaxies on the MS and quiescent galaxies below the MS. There is no observed relationship between X-ray luminosity and distance from the MS.

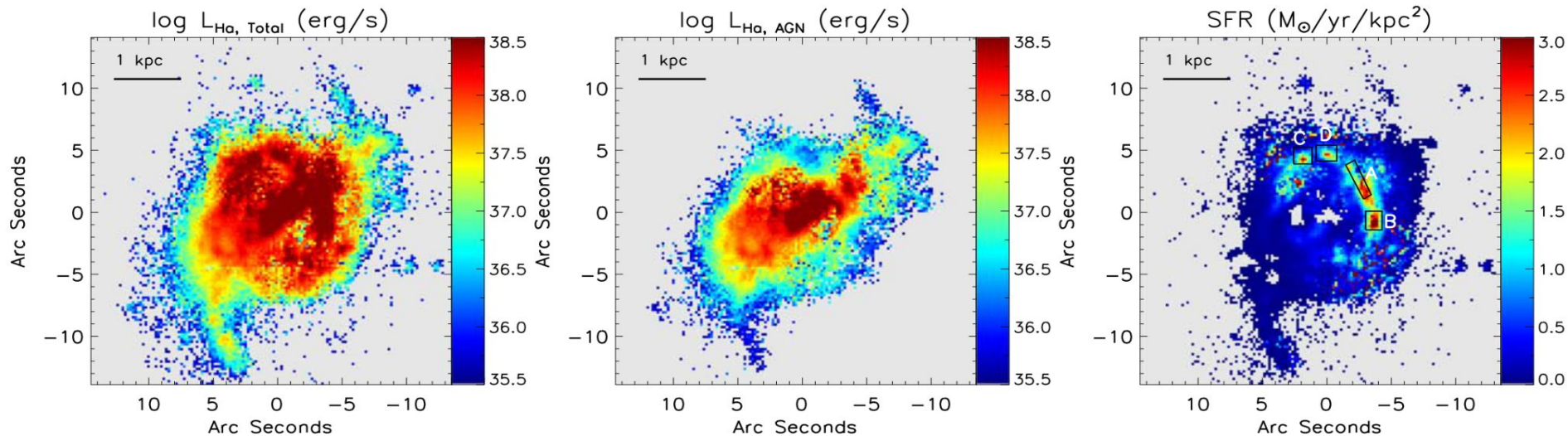


Positive Feedback

Observational Evidence - Shin et al. (2019)

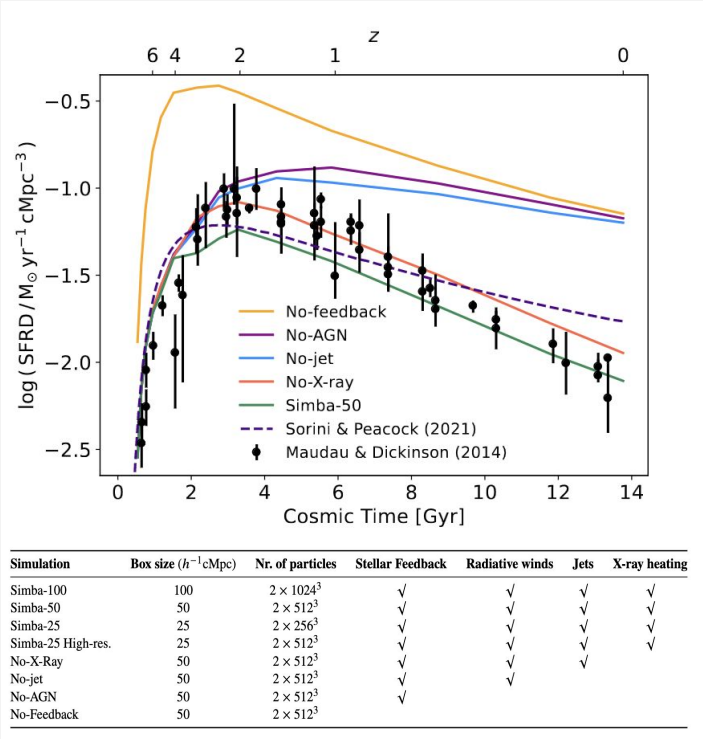
Shin et al. (2019) studied the Seyfert 2 galaxy, NGC 5728.

Findings: It was detected a high SFR in Reg.A, where the found gas outflows encounter the star formation ring. The star formation efficiency of Reg.A is higher than other regions in the ring by a factor of $\sim 3\text{--}5$, which suggests that the outflows may have enhanced SF in the area. The relative deficit of CO in Reg.A also indicates that the triggering mechanism of star formation is different compared to other regions.



Negative Feedback

Simulations - Scharré et al. (2024)



Scharré et al. (2024) investigated the impact of stellar and AGN feedback on the cosmic star formation history using multiple cosmological Simba simulation runs.

Findings: While AGN feedback starts quenching star formation in individual haloes after $z = 4$, $z = 2$ emerges as a key redshift transition, after which AGN feedback modes take over as the dominant quenching mechanisms. Before $z = 2$, the shape of global statistics like the SFRD is set by stellar feedback.

Among the three AGN feedback modes in Simba, jet feedback at low accretion rates is by far the most effective at quenching star formation.

Conclusions

AGN feedback plays a complex and dual role in galaxy evolution, capable of both quenching and stimulating star formation depending on different factors.

Observational studies and simulations consistently show AGN feedback's significant influence on star formation rates, with findings of both suppressed and enhanced star formation in AGN host galaxies. These mechanisms are crucial in shaping the star formation history and physical properties of galaxies, influencing their evolution over cosmic time. Thus, AGN feedback's diverse and dynamic effects are essential considerations in studies of galaxy evolution.

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